

Design Pattern 7: Observer Pattern

Exercise 7: Implementing the Observer Pattern

Scenario:

You are developing a stock market monitoring application where multiple clients need to be notified whenever stock prices change. Use the Observer Pattern to achieve this.

Steps:

Create a New Java Project:

Create a new Java project named ObserverPatternExample.

Define Subject Interface:

Create an interface Stock with methods to register, deregister, and notify observers.

Implement Concrete Subject:

Create a class StockMarket that implements Stock and maintains a list of observers.

Define Observer Interface:

Create an interface Observer with a method update().

Implement Concrete Observers:

Create classes MobileApp, WebApp that implement Observer.

Test the Observer Implementation:

Create a test class to demonstrate the registration and notification of observers.